University of Minho
School of Engineering

CENTROALGORITMI

Interoperability as the Foundation for Healthcare AI

Introducing openEHR



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Who am I?



Cristiana Neto

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Academic Background

2018 — MsC in Biomedical Engineering - Medical Informatics
University of Minho

2024 — PhD in Biomedical Engineering
University of Minho

Teaching Experience

Databases	BSc Computer Engineering
NoSQL Databases	MSc Computer Engineering
Biomedical IT Applications	MSc Biomedical Engineering
Knowledge Discovery	MSc Computer Engineering
Numerical Computing Languages	BSc Physics

Who am I?

Research Lines

openEHR & Interoperability

Standardising clinical records using openEHR.

Clinical AI & Decision Support

Prediction models for sepsis, readmission, chronic disease, and cancer using ML and data mining.

Hospital Workflow Automation

Automation of imaging exam workflows using international standards. Published in CMPB 2025.

University of Minho

Est. 1973 · Braga & Guimarães, Portugal



- **Top 100 universities < 50 yrs** (*76th position*)

Braga – Gualtar Campus





Guimarães – Azurém Campus





21,000+	1,033	10	31
students	academic staff	schools / institutes	research labs

- School of Architecture (EA)
- School of Letters, Arts and Human Sciences (ELACH)
- School of Psychology (EP)
- Institute of Education (IE)
- Institute of Social Sciences (ICS)
- School of Economics and Management (EEG)
- School of Engineering (EE)
- School of Law (ED)
- School of Sciences (EC)
- School of Medicine (EM)

Centro ALGORITMI — University of Minho

R&D unit in ICT & Electronics · School of Engineering

6 Major Research Fields

CST – Computer Science & Technology
IST – Information Systems & Technology
CCPM – Computer Communications & Pervasive Media
IE – Industrial Electronics
IEM – Industrial Engineering & Management
SEOR – Systems Engineering & Operational Research

5 Thematic Lines

- Innovative & Sustainable Industries
- Smart Cities, Energy & Mobility
- Health & Well-being
- Infrastructures & Connected Society
- Public Administration & Government

540+ researchers

4 departments

LASI associated lab

CENTROALGORITMI

KEG — Knowledge Engineering Group is one of the research groups within ALGORITMI (member since 2018)

Lecture Agenda

1. Contextualization
2. Interoperability
3. Introducing openEHR & archetypes
4. Modelling using openEHR
5. openEHR vs FHIR
6. Conclusions

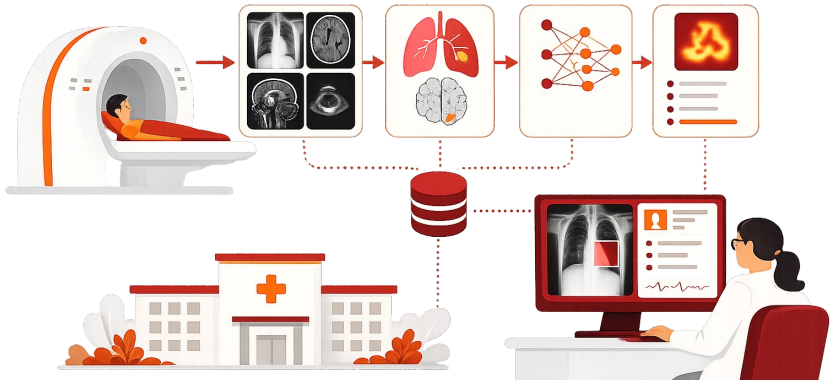
Context

You already know

- Medical imaging (X-ray, CT, MRI, US)
- Visualization & segmentation
- AI pattern detection in images
- Basic CNN / deep learning architectures

Today we will discuss

- What happens when AI enters a real hospital
- Why images alone are not enough
- How openEHR gives AI the context it needs



Starting Point

AI outputs from images

Classification
"Pneumonia detected — 92% confidence"

Detection
Bounding boxes on nodules or lesions

Segmentation
Pixel-level organ / tumour masks

Heatmaps
Regions driving the prediction

Regression
Bone age estimation, tumour volume

A question NOT answered

"Should we give this patient antibiotics right now?"

This requires context beyond the image.

Key Limitation

A chest X-ray finding of pneumonia is just the starting point. The clinical decision needs at least five more dimensions.

AI Finding

Pneumonia detected

94% confidence

X-ray only

But also needs …

Symptoms	*Fever? Cough? Since when?*
Vital signs	*SpO_2, HR, RR, temperature*
Patient history	*Comorbidities, prior pneumonias, allergies*
Lab results	*WBC, CRP, procalcitonin, cultures*
Medications	*Already on antibiotics?*

Key Challenges

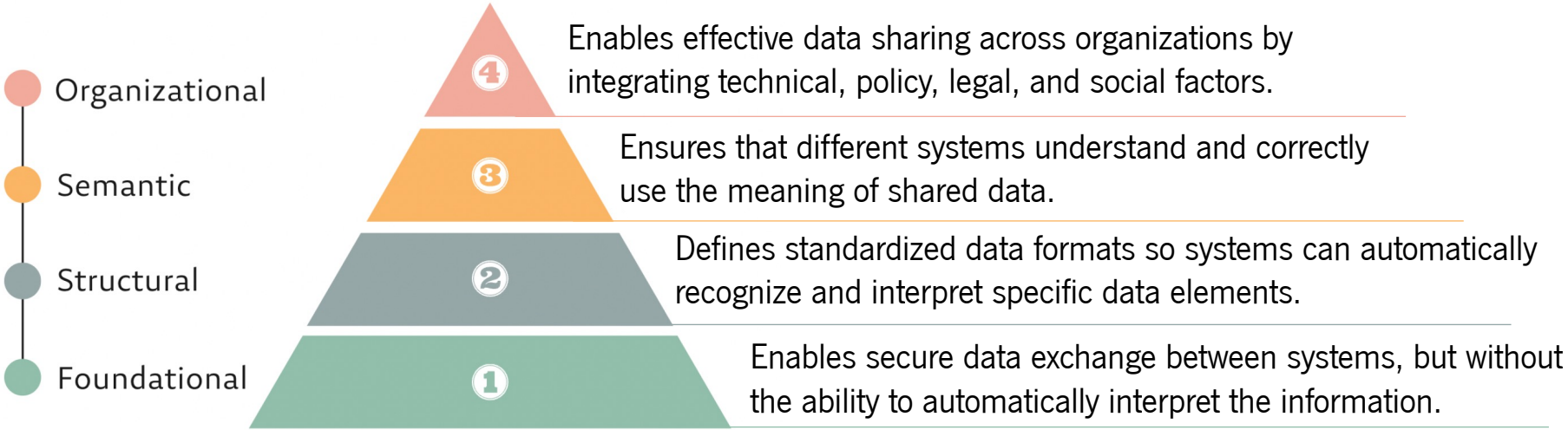
47%

annual growth rate of healthcare data volume

- **Data access** — information dispersed across EHRs, HIEs, and unaffiliated providers
- **Data quality** — incomplete or incorrect entries amplify AI errors and biases
- **Heterogeneity** of formats across hospitals, regions and community partners
- **SDOH & behavioral health integration** — no longer optional for whole-person care
- **Regulatory shifts** — TEFCA/QHIN participation raising the bar for data exchange
- **Governance** — consent, privacy, auditability required for responsible AI adoption

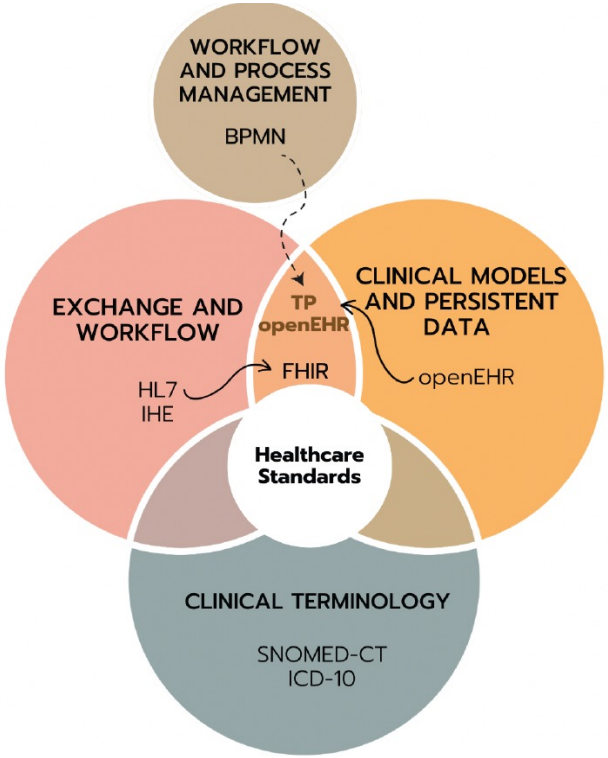
Interoperability

Interoperability refers to the methods, protocols, terminologies and specifications for the collection, exchange, storage and retrieval of health information.



Interoperability Standards

Healthcare interoperability standards enable seamless data exchange across systems and organizations, ensuring data can be accessed, understood, and used by providers, researchers, and patients.



Key standards in healthcare interoperability:

- openEHR
  - support data structure
- HL7, and FHIR
  - support data exchange
- ICD-10 and SNOMED-CT
  - provide standardized clinical terminology

Introducing openEHR

openEHR is a set of open specifications, clinical models, and software that creates standards for healthcare IT, solving one of the sector's biggest challenges: interoperability.

- Vendor-neutral standard;
- Highlights the semantics, definition and structure of clinical data;
- Collaborative environment that incorporates healthcare professionals;
- Improves coordination and consistency in care processes.

Introducing openEHR

openEHR

01	**Specifications**	The technical foundation: models and languages that make everything work together.
02	**Clinical Modeling**	Standardised ways to record clinical information (archetypes) reusable across systems.
03	**Software**	Open-source tools and components to build interoperable EHRs.
04	**Education**	Building communities and sharing best practice to implement openEHR globally.

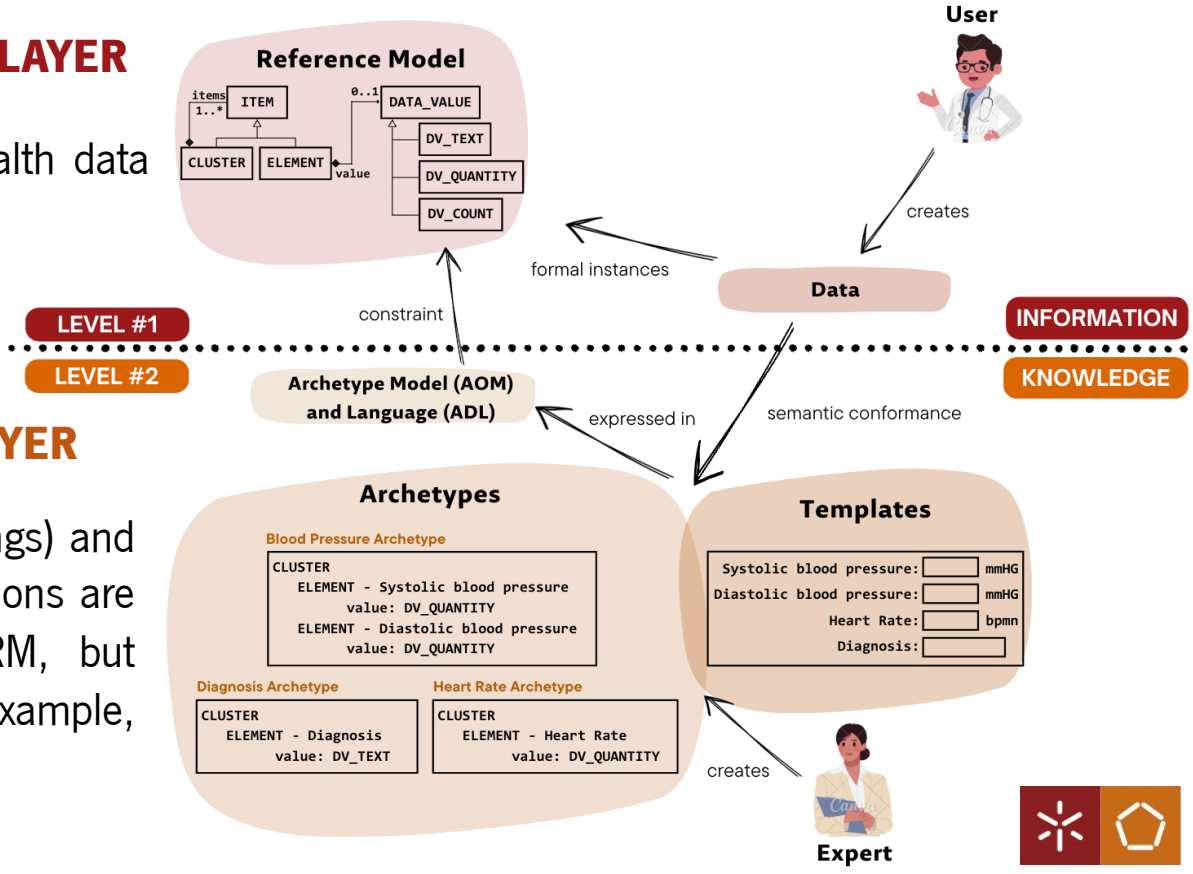
openEHR Dual-model Approach

Level 1 (UML) – INFORMATION LAYER

Reference Model (RM) – how health data are represented in the patient record.

Level 2 (ADL) – KNOWLEDGE LAYER

Archetypes (with terminology bindings) and **Templates** – clinical content definitions are represented separately from the RM, but automatically inherit parts of it (for example, data types).



openEHR: Basics

1

Archetypes



Specification of reusable clinical concepts

standardized content

2

Templates



Combination of multiple archetypes to serve a specific use case

case-specific content

3

Forms



Defines the graphical interface based on the template

case-specific functionality

openEHR: Archetypes

They are computable specifications for a single clinical concept or instrument:

- Blood pressure
- Diagnosis
- NEWS2
- Symptoms
- Laboratory results
- etc.





Objective:

To aggregate as much data as possible for a clinical concept in a generic way.

Method:

Build once and reuse whenever needed, depending on the use case.

Archetypes are built on top of RM classes and inherit their attributes.
Governance of these models is crucial.

openEHR: Templates

A template is a special type of archetype:
It acts as a “parent” archetype that aggregates multiple “child” archetypes.

Objective:
To address specific use cases.



Governance:
It does not require governance as strict as archetypes.



Note:
Typically, building a template requires an archetype of type **COMPOSITION**.

openEHR: Templates

Archetypes are building blocks. Templates assemble them into a complete clinical document for a specific use case.

Reference Model
openEHR RM — the fixed technical skeleton

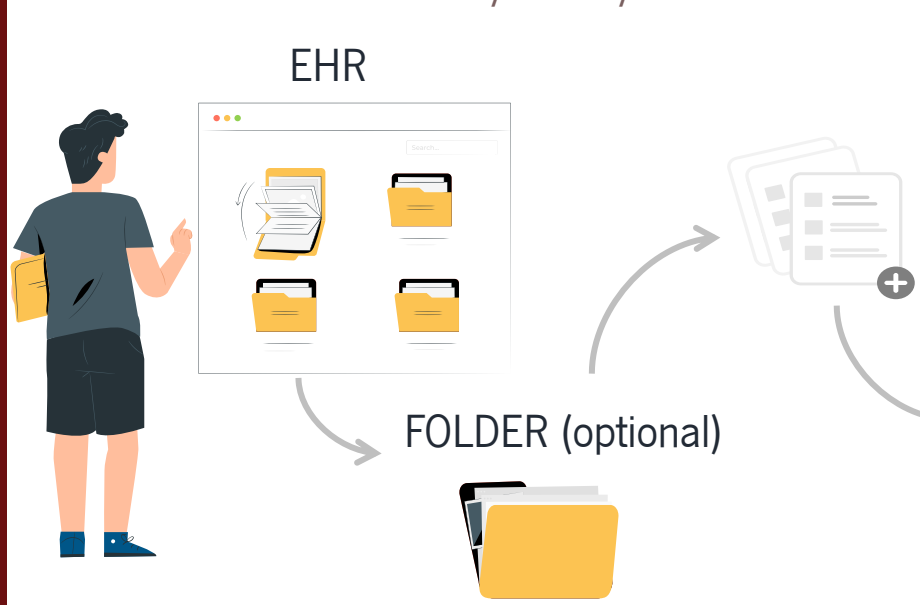
Archetypes
Reusable clinical concept definitions — shared globally

Templates
Constrained combination of archetypes for a specific context

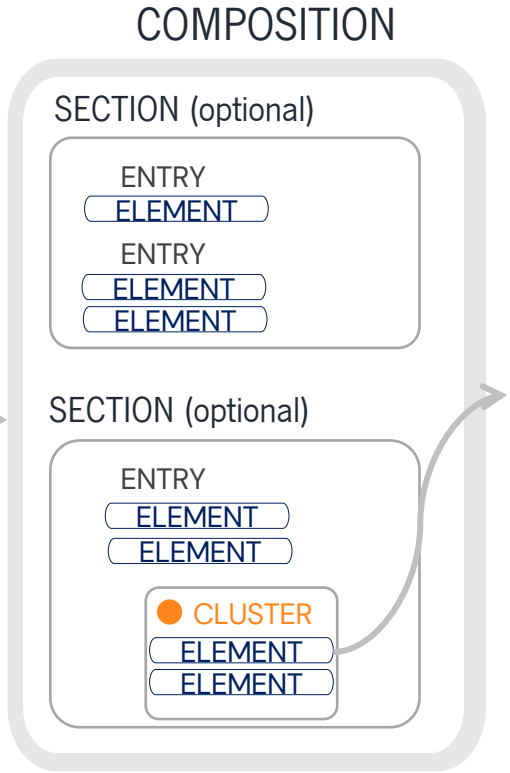
Composition (instance)
Actual patient data — one completed form

openEHR: Record Structure

O EHR inclui uma hierarquia de pastas



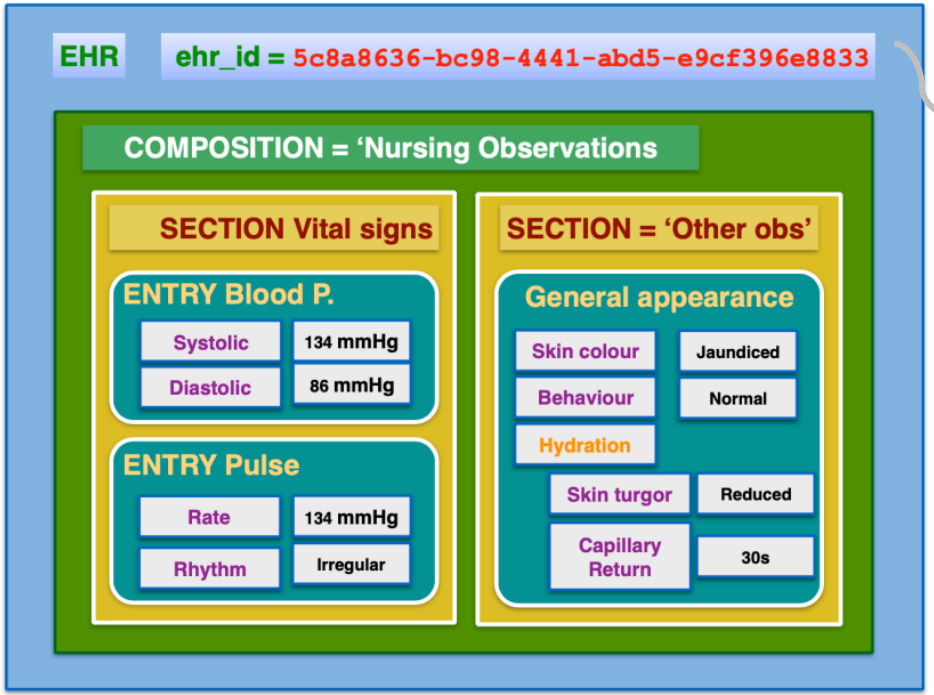
Each folder containing multiple compositions



- DV_BOOLEAN
- DV_TEXT
- DV_CODED_TEXT
- DV_ORDINAL
- DV_COUNT
- DV_PROPORTION
- DV_QUANTITY
- DV_IDENTIFIER
- DV_DATE
- DV_TIME
- DV_DATE_TIME
- ...

openEHR: Record Structure

openEHR Data Objects



- Top-level container for all of the data for a **single patient**
- Each EHR has a unique, anonymous ID the `ehr_id`
- This needs to be associated with a real-world identifier e.g NHS Number to allow the patient to be identified

openEHR: Composition

- **Definition:** It is the committal unit of data, analogous to a document.
- **Structure:** Compositions are defined by Templates (which restrict archetypes) and are instances of the openEHR Reference Model COMPOSITION class.
- **Context:** Contains essential context about the data, including category (e.g., persistent or event), time, and subject.

Common Composition Archetypes



Encounter
`openEHR-EHR-COMPOSITION.encounter.v1`



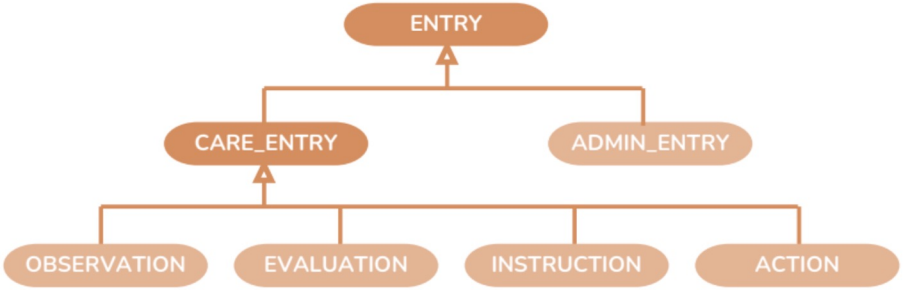
Report
`openEHR-EHR-COMPOSITION.report.v1`



Request for servisse
`openEHR-EHR-COMPOSITION.request.v1`

openEHR: Entry

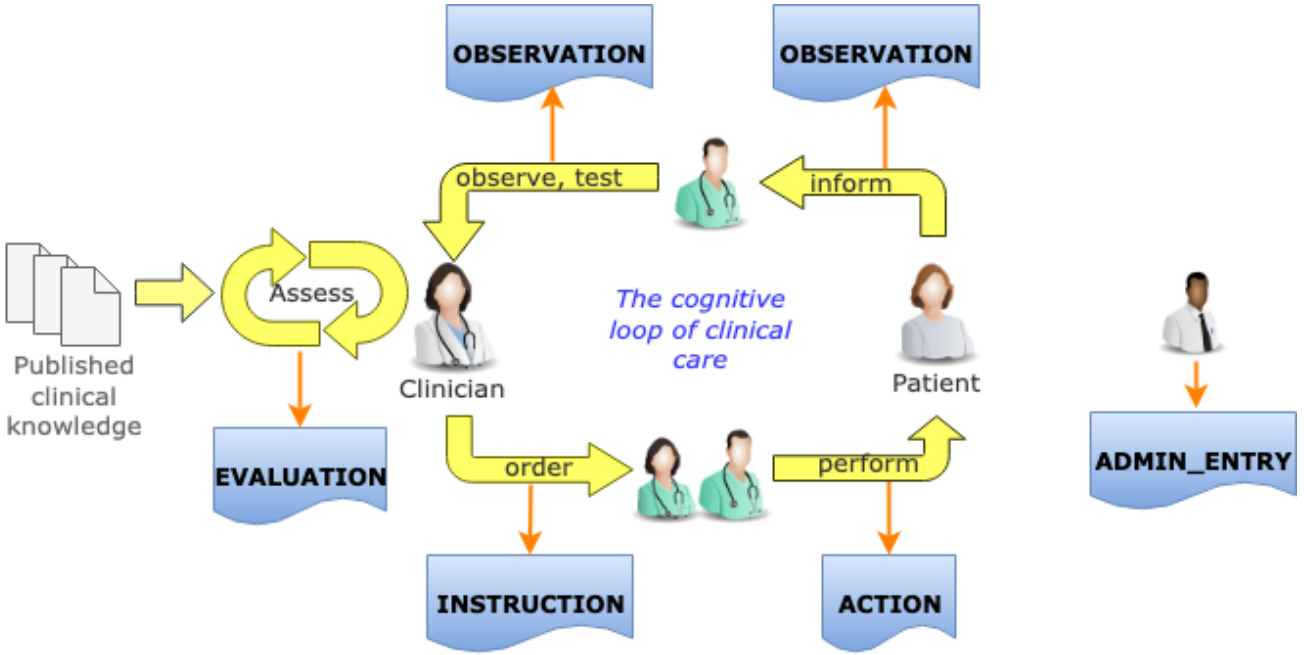
Contains the generic structures for recording clinical content.



- **ADMIN_ENTRY:** for administrative information.
- **OBSERVATION:** for observable phenomena, including mechanical or manual measurements, such as temperature, and responses to interviews.
- **EVALUATION:** for clinical opinions such as assessments, diagnoses, and plans.
- **INSTRUCTION:** to record a set of rules and actionable instructions, such as medication requests, monitoring, and reviews, including cancellation or postponement. It will have an **ACTION** as a response.
- **ACTION:** information recorded as a result of carrying out a step of a clinical activity, including cancellation or postponement. It may be a response to an **INSTRUCTION**.

openEHR: EHR Information Model

Overview



Source: https://specifications.openehr.org/releases/RM/latest/ehr.html

openEHR: AQL

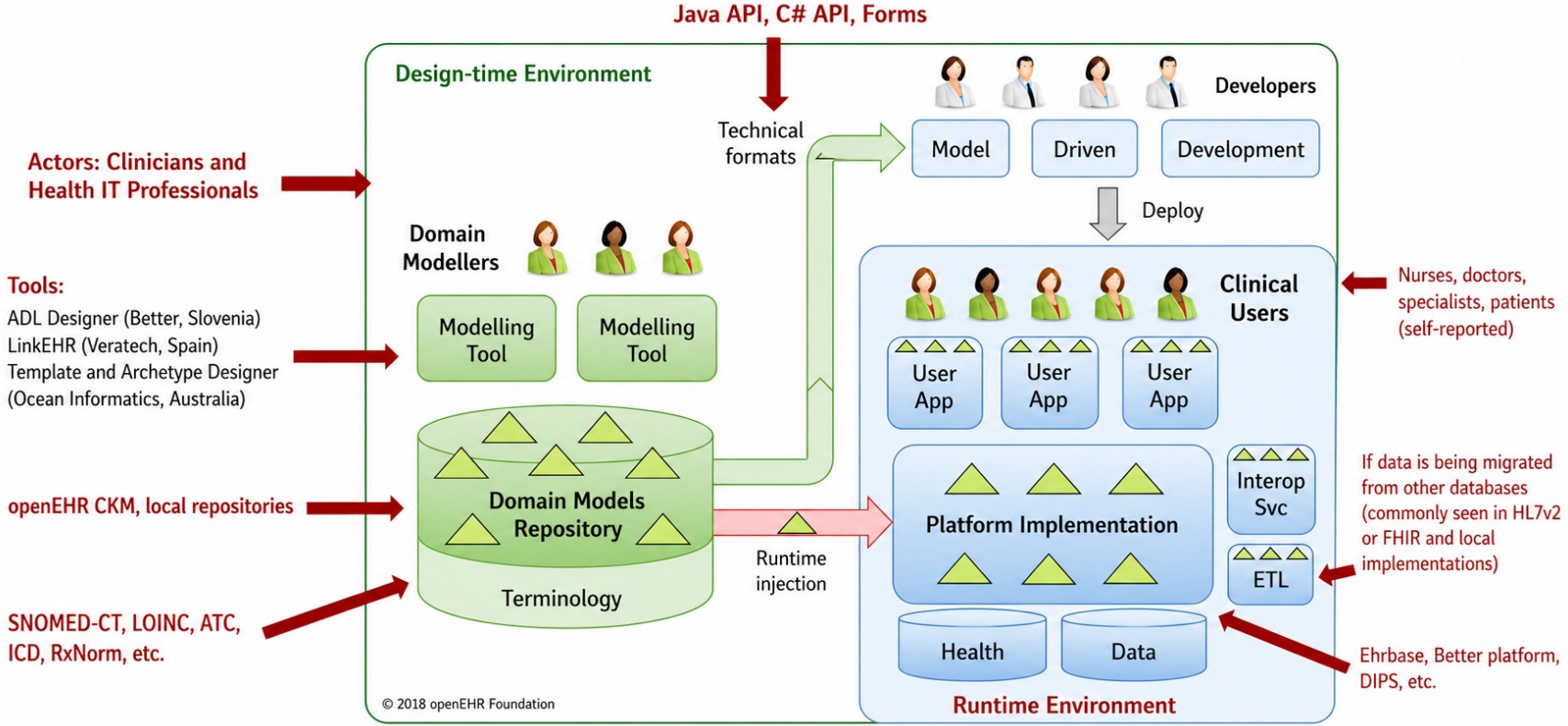
AQL – Archetype Query Language

- Query language for retrieving data from openEHR systems
- Similar to SQL, but designed for clinical data structures
- Works on archetypes and paths (e.g., /data/events/...), not database tables
- Queries COMPOSITION, EHR, ENTRY structures

Example: Query to retrieve blood pressure measurements for a patient and identify hypertension cases (≥140/90 mmHg).

```
SELECT
   o/data[at0001]/events[at0006]/data[at0003]/items[at0004]/value/magnitude,
   o/data[at0001]/events[at0006]/data[at0003]/items[at0005]/value/magnitude
FROM
   EHR [ehr_id/value='1234']
      CONTAINS COMPOSITION [openEHR-EHR-COMPOSITION.encounter.v1]
         CONTAINS OBSERVATION o [openEHR-EHR-OBSERVATION.blood_pressure.v1]
WHERE
   o/data[at0001]/events[at0006]/data[at0003]/items[at0004]/value/magnitude >= 140 OR
   o/data[at0001]/events[at0006]/data[at0003]/items[at0005]/value/magnitude >= 90
```

openEHR Architecture



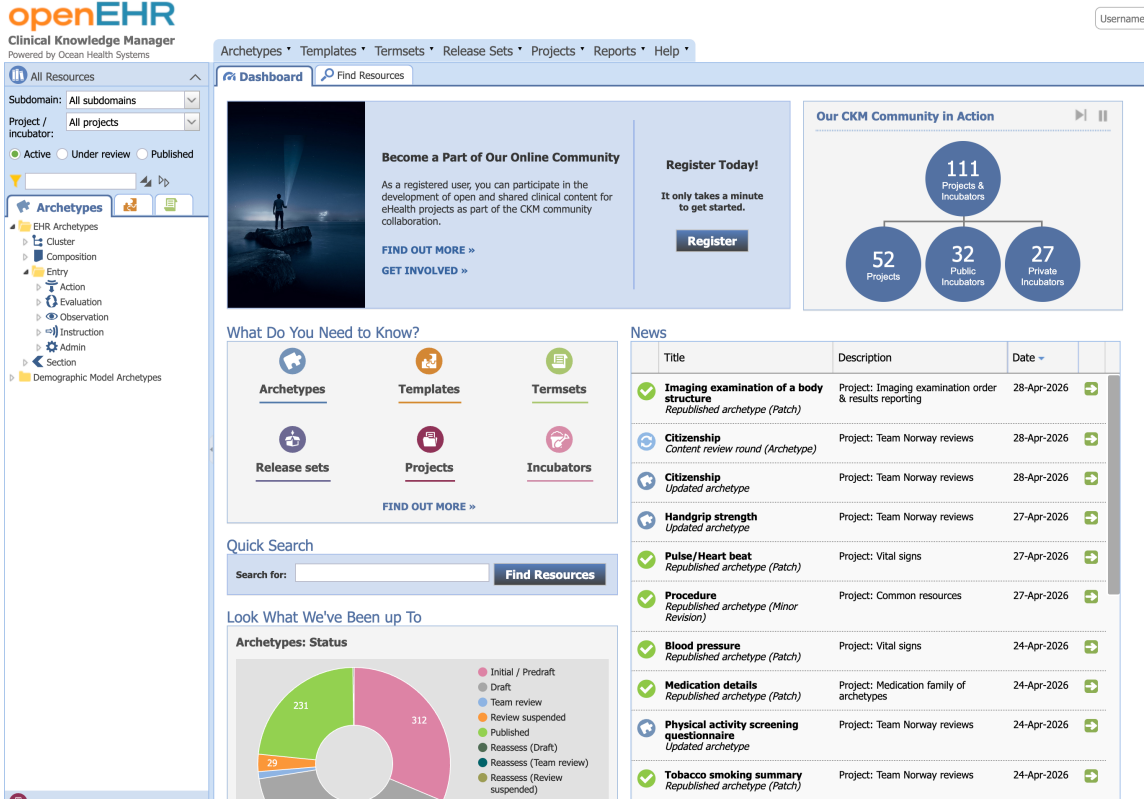
The model-driven openEHR technology ecosystem

Source: https://openehr.org/developer-ecosystem/

openEHR: Tools

Clinical Knowledge Manager (CKM)

https://ckm.openehr.org/ckm/

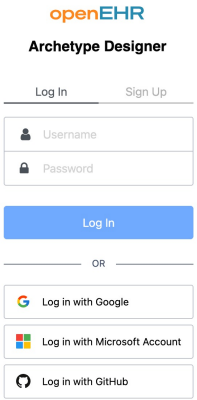


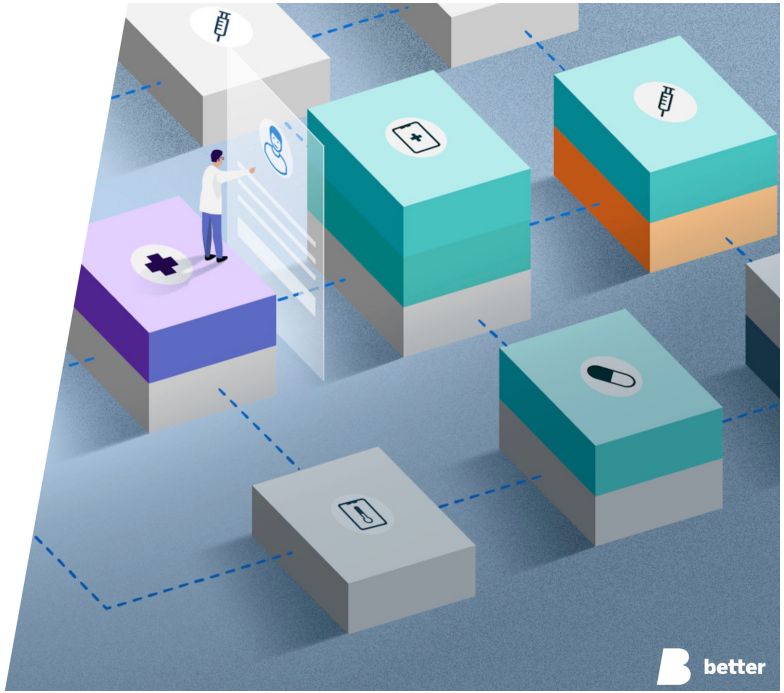
The CKM is a repository of the main pillars of openEHR, as it enables global sharing and collaboration in the creation and improvement of archetypes.

openEHR: Tools

Archetype Designer (AD)

https://tools.openehr.org/designer/





A free web service that provides a clinical modeling environment for the development of openEHR archetypes, enabling the “visual creation” of archetypes and templates, including the analysis, validation, and full serialization of archetypes.

Example: Lung nodule segmentation

How do we clinically represent an AI result in medical imaging?

More Traditional Approach	Structured OpenEHR
• Free-text in the report • “A 12mm nodule was found” • No structure ⟶ not searchable • No link to DICOM	• Structured finding type • Numeric value: 12 mm (DV_QUANTITY) • Searchable, aggregable, comparable • DICOM reference (Study UID / SEG UID)

Example: Lung nodule segmentation

Goal: structured version of this report

Radiology Report — CT Chest

Patient	M.C., 58 y/o, male
Date	15 Mar 2024
Exam	CT Chest (with contrast)
Referred by	Dr. J. Silva — Pulmonology
Indication	Persistent cough, ex-smoker

Findings: There is a small nodule in the right lung, upper area, roughly 1.2 cm, irregular borders. The segmentation tool (LungSegNet v2.1) calculated a volume of 12 mm3 — confidence 0.87, which seems reliable. Rest of the lung looks mostly ok, some mild emphysema changes bilaterally. No pleural effusion.

Conclusion: Suspicious nodule RUL. Follow-up recommended. Emphysema.

Dr. Sarah Owen
Radiologist



Example: Lung nodule segmentation

Step 1: Choosing the Base Archetypes

Search in CKM and reuse: you shoud never start from scratch in OpenEHR

OBSERVATION.imaging_examination_result
Imaging exam result (CT, X-Ray, MRI...)
CKM ↗

Structured imagens findings:

CLUSTER.imaging_exam_anomaly.v0
Individual anomaly finding — nodule, lesion, structure
CKM ↗

Physical properties:

CLUSTER.physical_dimensions.v1
Spatial measurements of a structure
CKM ↗

Additional details:

CLUSTER.device.v1
Instrument in the provision of healthcare
CKM ↗

Example: Lung nodule segmentation

Step 2: Template Creation

- A template always defines a clinical document.
- In openEHR, clinical documents are represented as COMPOSITIONs.
- Before building the template, we need to decide what kind of document we are modeling.

openEHR-EHR-COMPOSITION.report-result.v1
Document to communicate information to others about the result of a test or assessment.

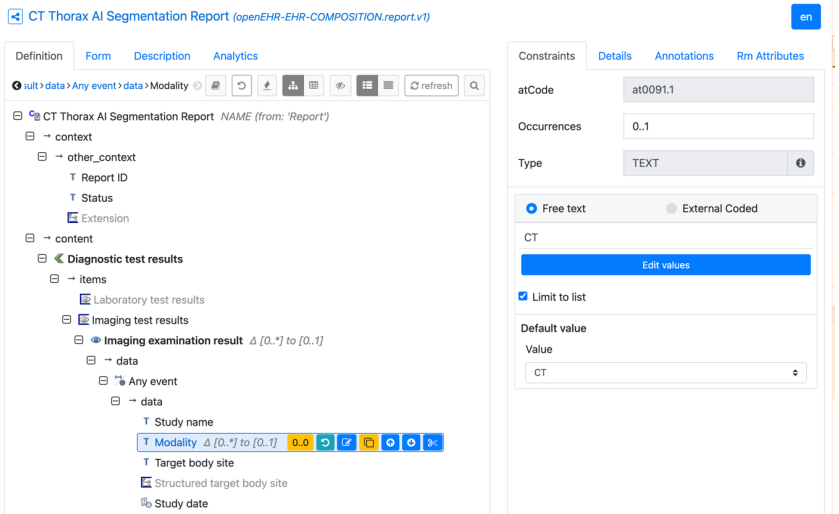
CKM ↗

Example: Lung nodule segmentation

Step 3: Template Creation

- Combine archetypes into a coherent structure for a specific clinical use case;
- Define how archetypes are organized;
- Restrict and simplify data elements;
- Adapt the model to the clinical scenario.

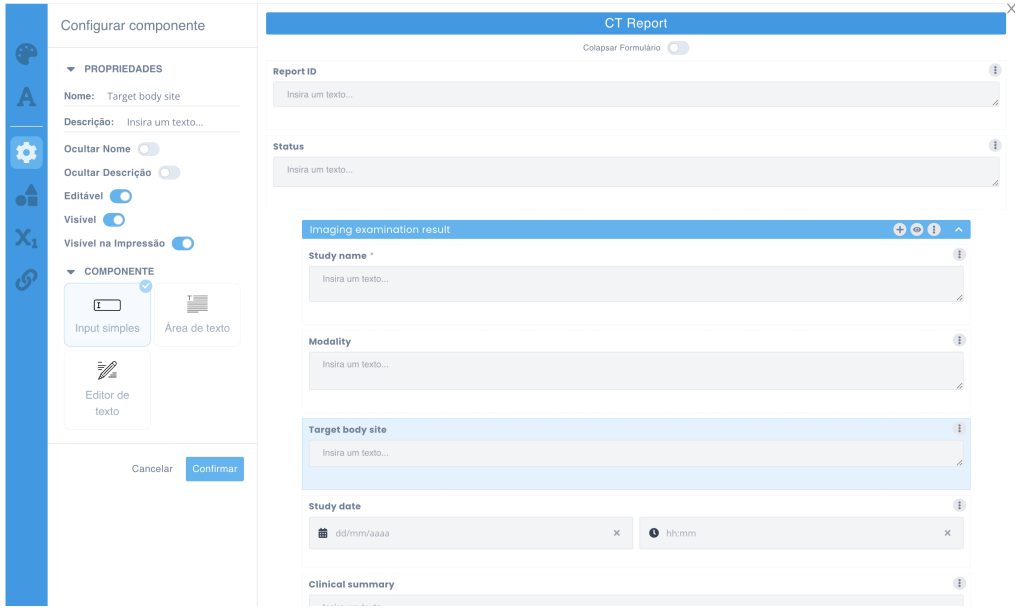
Done in the Archetype Designer tool



Example: Lung nodule segmentation

Step 5: Form Creation

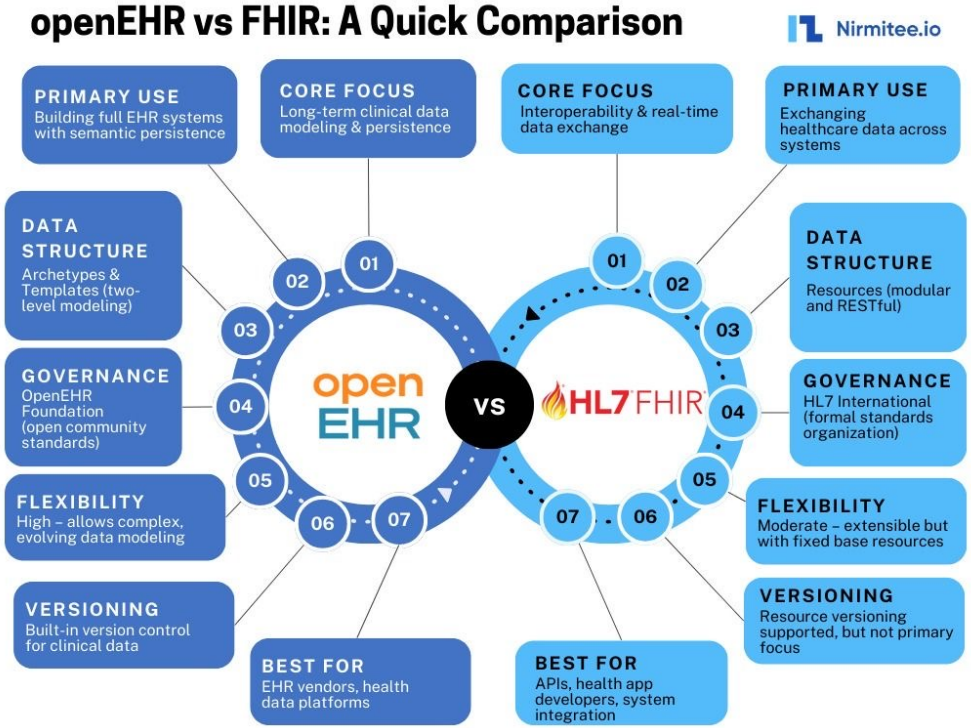
Create a user-facing form/UI based on the template.



...

openEHR vs FHIR

Competition or Complementarity?



If we combine both, we move from exchanging data...

to truly understanding it.

Source: https://www.linkedin.com/posts/nirmitee_healthtech-fhir-openehr-activity-7325134687692046338-R_al/

Conclusions

- **Images alone are not enough** — AI outputs gain more clinical value when combined with structured patient context.
- **Interoperability is very important for responsible AI**
- **openEHR separates technical and clinical concerns** — the RM is a stable foundation; archetypes capture clinical knowledge independently.
- **openEHR and FHIR are complementary** — openEHR for longitudinal storage, FHIR for real-time exchange.

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CENTROALGORITMI

Thank you!

Questions?

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